JC2066: IT Professionals: Ethical, Legal and Social Issues

L01: Artificial Intelligence

Jan. 16th, 2026

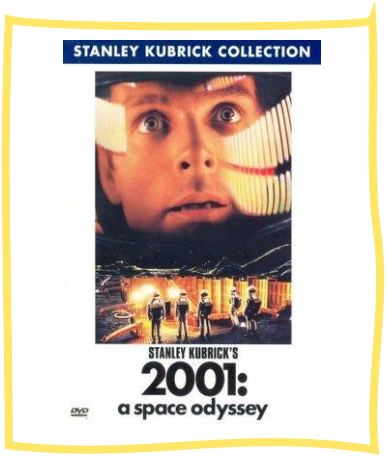
MA, Jiawei Phoenix
Department of Computer Science

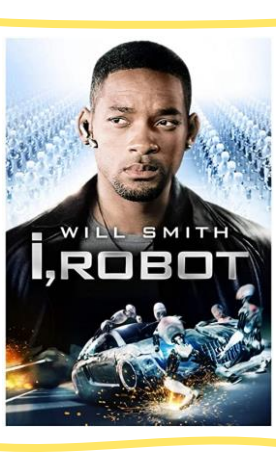
Where are the rest of the resources for this class?
On Canvas: https://canvas.cityu.edu.hk/courses/68163

Today’s Agenda

1. **What can AI do?**
2. What is AI?
3. AI and Machine Learning

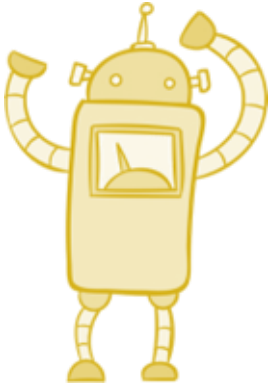
What Can AI Do?





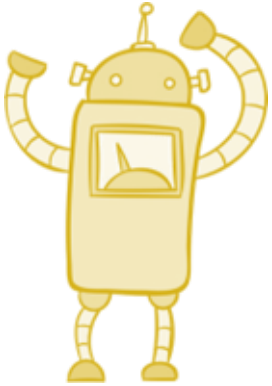
What Can AI Do?

Play a game of badminton.
Play a game of jeopardy.
Drive safely along a curving mountain road.
Drive safely along Nathan Road.
Buy a week’s worth of food on the web.
Buy a week’s worth of food at Parken Shop.
Discover and prove a new mathematical theorem.
Chat succesfully with another person for an hour.
Perform a surgical operation.
Translate spoken Chinese into spoken English in real time.
Write an intentially funning story.



What Can AI Do?

- Play a game of badminton.
- Play a game of jeopardy.
- Drive safely along a curving mountain road.
- Drive safely along Nathan Road.
- Buy a week’s worth of food on the web.
- Buy a week’s worth of food at Parken Shop.
- Discover and prove a new mathematical theorem.
- Chat succesfully with another person for an hour.
- Perform a surgical operation.
- Translate spoken Chinese into spoken English in real time.
- Write an intentially funning story.



Nature Language Processing

Question answering

Context: California is a major economic center for US…

Question: What is a major importance of California?

Anaswer: major economic center

Machine translation

French: Cette rubrique comprend plusieurs tâches connexes.

English: This rubric includes several related tasks.

Sentiment analysis

Resturant review: The dessert is excellent.

Sentiment: Positive

Web search

Query: CityU CS Artificial Intelligence

Top-2 results:

www.cityu.edu.hk › current › course ▾ 翻译此页

CS4486 - Artificial Intelligence - City University of Hong Kong

CS4486 - Artificial Intelligence ... Department of Computer Science ... skills of problem solving

www.cs.cityu.edu.hk › research_areas › Artifi... ▾ 翻译此页

Artificial Intelligence And Knowledge Management - CityU CS

Artificial Intelligence And Knowledge Management - Research Areas - Department of Computer

Nature Language Processing

Speech

Automatic speech recognition

Speech:

Text: This rubric includes several related tasks.

Text-to-speech synthesis

Text: This rubric includes several related tasks.

Speech:

Chatbot

User: Are you mad at me?

Siri: I can't answer that.

User: Do you love me?

Siri: Would you like me to search the web for "love"?

Speaker verification

Speech:

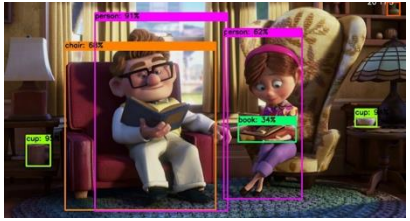
Question: Is this Bob's voice?

Answer: Yes

Speech

Perception

Object and face recognition



Scene segmentation

Image classification

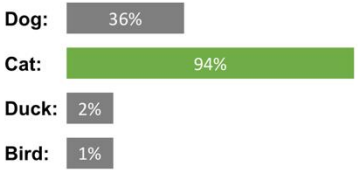


Image captioning

Perception

Robotics

- Self-Driving Car
- Warehousing robot
- Rescue robot
- Soccer cobots

Self-driving Car

Warehousing Robot

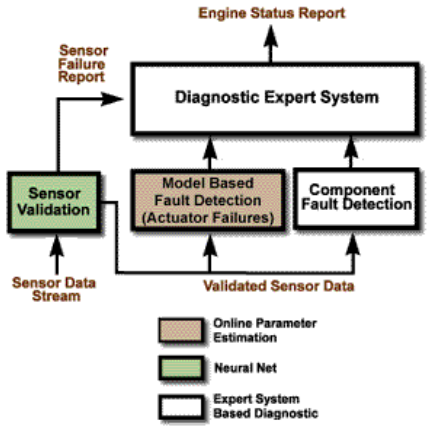
Logic

Theorem provers

Machine proof in LT

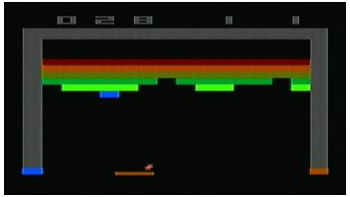
*2.85 $p \vee q . \supset . p \vee r : \supset : p . \vee . q \supset r$	
1. $p \supset q . \supset : q \supset r . \supset . p \supset r$	*2.06
$p/q, q/p \vee q, r/p \vee r$	
2. $q \supset p \vee q . \supset : p \vee q \supset p \vee r . \supset . q \supset p \vee r$	Sub. 1
3. $q \supset p \vee q$	*1.3
4. $p \vee q \supset p \vee r . \supset . q \supset p \vee r$	*Modus ponens* 3, 2
5. $p \vee q \supset p \vee r . \supset . \sim q \vee (p \vee r)$	Def. of $\supset$
6. $p \vee (q \vee r) . \supset . q \vee (p \vee r)$	*1.5
$p/\sim q, q/p$	
7. $\sim q \vee (p \vee r) . \supset . p \vee (\sim q \vee r)$	Sub. 6
8. $p \vee q \supset p \vee r . \supset . p \vee (\sim q \vee r)$	Chain with 5 and 7
9. $p \vee q \supset p \vee r . \supset . p . \vee . q \supset r$	Def of $\supset$

NASA fault diagnosis



Game Playing

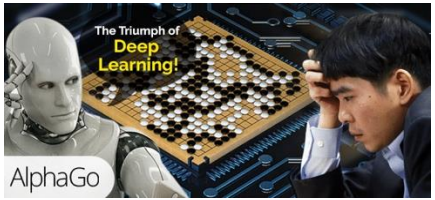
Chess



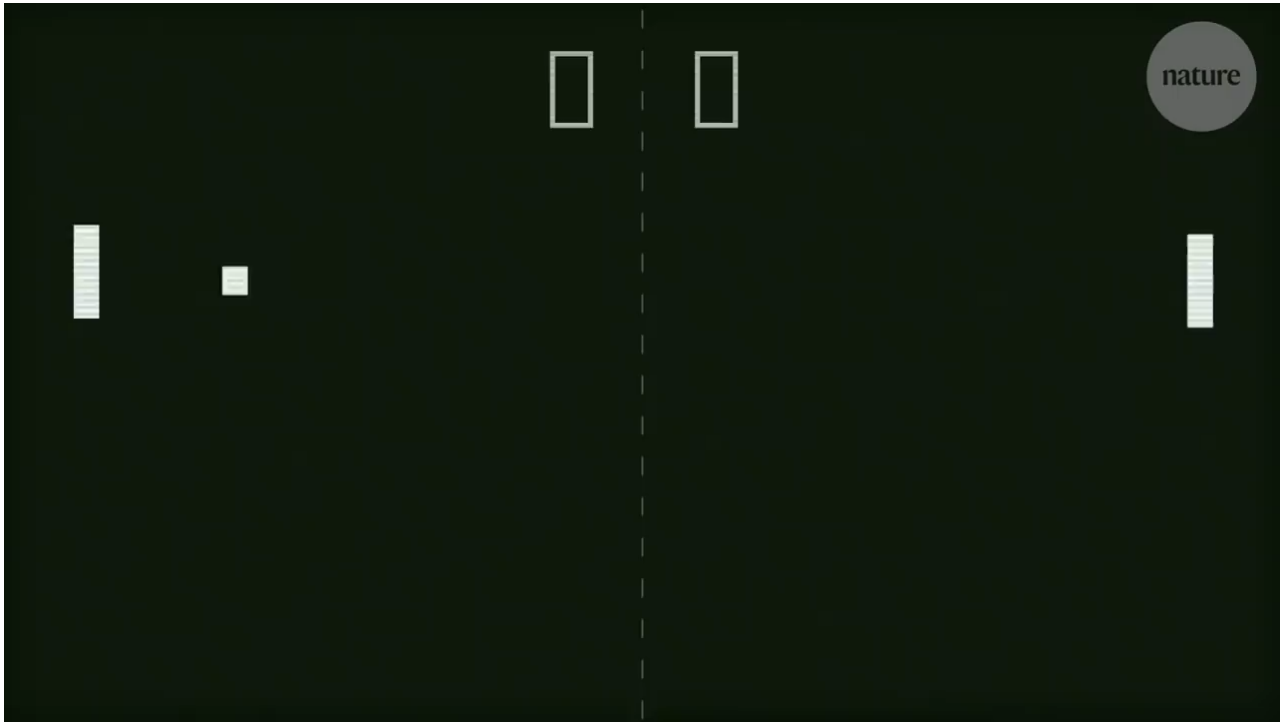
Atari games

Jeopardy

Go



Game Playing

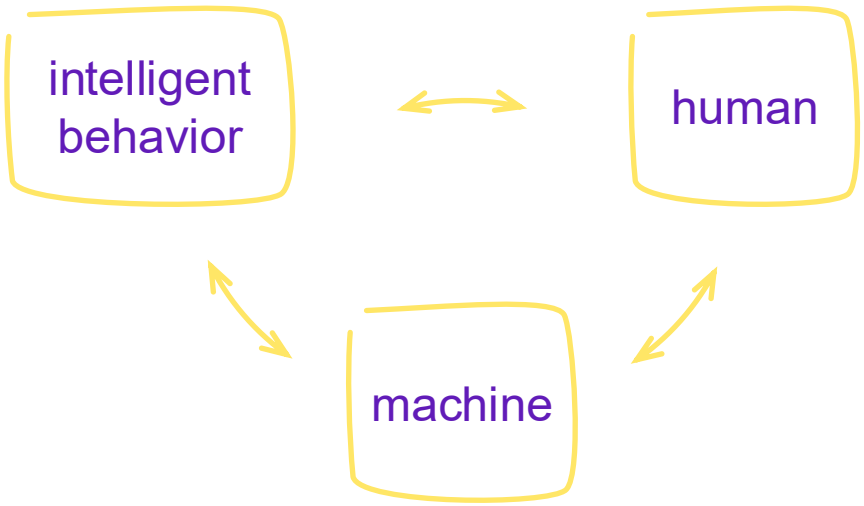


Today’s Agenda

1. What can AI do?
2. **What is AI?**
3. AI and Machine Learning

What is AI?

- AI is an attempt of reproduction of human reasoning and intelligent behavior by computational methods.



What is AI?

- [Russell & Norvig] AI is a discipline that systematizes and automates reasoning processes to create machines that

	like humans	rationally
think	cognitive science	laws of thought
act	Turing test	rational agents

What is AI?

- [Russell & Norvig] AI is a discipline that systematizes and automates reasoning processes to create machines that

	like humans	rationally
think	cognitive science	laws of thought
act	Turing test	rational agents

Thinking Humanly

Modelling how humans think

- Introspection (too subjective)
- Psychological experiments (predicting behaviors of humans based on cognitive theory and then testing, with matched predictions as supporting evidence)
- Brain imaging (observing the brain in action)

Cognitive science

- Top-down: Computer models from AI to simulate a theory and experimental techniques from psychology
- Bottom-up: Neuropsysiological evidence incorporated into computational models

Thinking Humanly

- Cognitive science and AI are distinct disciplines, except the close connection of cognitive neuroscience to AI in neural networks.

- It is especially challenging to model how humans think, as brains are not as modular as software.

What is AI?

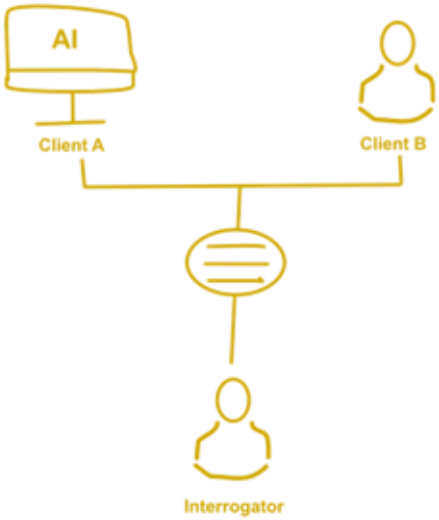
[Russell & Norvig] AI is a discipline that systematizes and automates reasoning processes to create machines that

	like humans	rationally
think	cognitive science	laws of thought
act	Turing test	rational agents

Acting Humanly

Turing test

- Proposed by Alan Turing in 1950
- "Can machines think?" → "Can machines behave intelligently?"
- Operational definition of intelligent behavior
- Can a human interrogator tell whether (written) responses to her (written) questions come from a human or an AI machine?
- Required capabilities: natural language processing, knowledge representation, automated reasoning, learning…



Acting Humanly

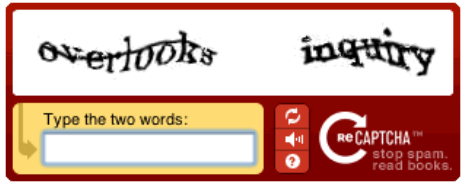
Attempts on passing the Turing test

- 1960s: ELIZA (by Joseph Weizenbaum)
- 1990s: ALICE
- Yearly: Loebner prize (http://www.loebner.net/Prizef/loebner-prize.html)

No machines has fully passed the test yet.

Applications of the Turing test

- CAPTCHA: Completely Automatic Public Turing tests to tell Computers and Humans Apart



Why do we want to replicate human behavior, including the imperfections?

Rationality

- An abstract ideal of intelligence — do the right thing rather than “whatever humans think or act”.

Definitions of “right”

- Logic: Conclusions are provable from inputs.
- Economics: The utility of outcomes is maximized.

What is AI?

- [Russell & Norvig] AI is a discipline that systematizes and automates reasoning processes to create machines that

	like humans	rationally
think	cognitive science	laws of thought
act	Turing test	rational agents

Thinking Rationally

Logicist tradition

- Syllogisms by ancient Greeks: argument structures that always yield correct conclusions given correct premises
- Logic: notation and rules for derivation for thoughts

Limitations

- Not all intelligent behavior is mediated by logical deliberation.
- It is difficult to represent informal knowledge in logic.
- Logical systems tend to do the wrong thing in the presence of uncertainty.

What is AI?

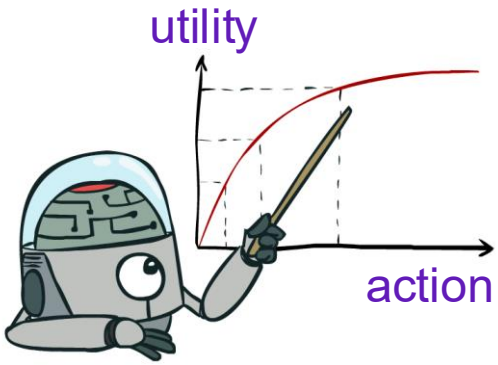
- [Russell & Norvig] AI is a discipline that systematizes and automates reasoning processes to create machines that

	like humans	rationally
think	cognitive science	laws of thought
act	Turing test	rational agents

Acting Rationally

- Rationality concerns what actions are made instead of the thought process behind them.

- Getting machines to be rational means maximizing the expected utility based on their circumstances and what they know.



Irrationally ≠ insane ; an irrationally solution could be sub-optimal.

Rationally ≠ successful ; the most rational action may fail due to out-of-control circumstances or due to incomplete knowledge.

A Short History of AI

1943: McCulloch & Pitts: Boolean circuit model of brain
1950: Turing's “Computing Machinery and Intelligence”

1969—79: Early development of knowledge-based systems
1980—88: Expert systems industry booms
1988—93: Expert systems industry busts: “AI Winter”

1940-50
Early days

1950-70
Excitement: Look, Ma, no hands

1970-90
Knowledge-based

1990-
Statistical

2000-
Large datasets

1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
1956: Dartmouth meeting: “Artificial Intelligence” adopted
1965: Robinson's complete algorithm for logical reasoning

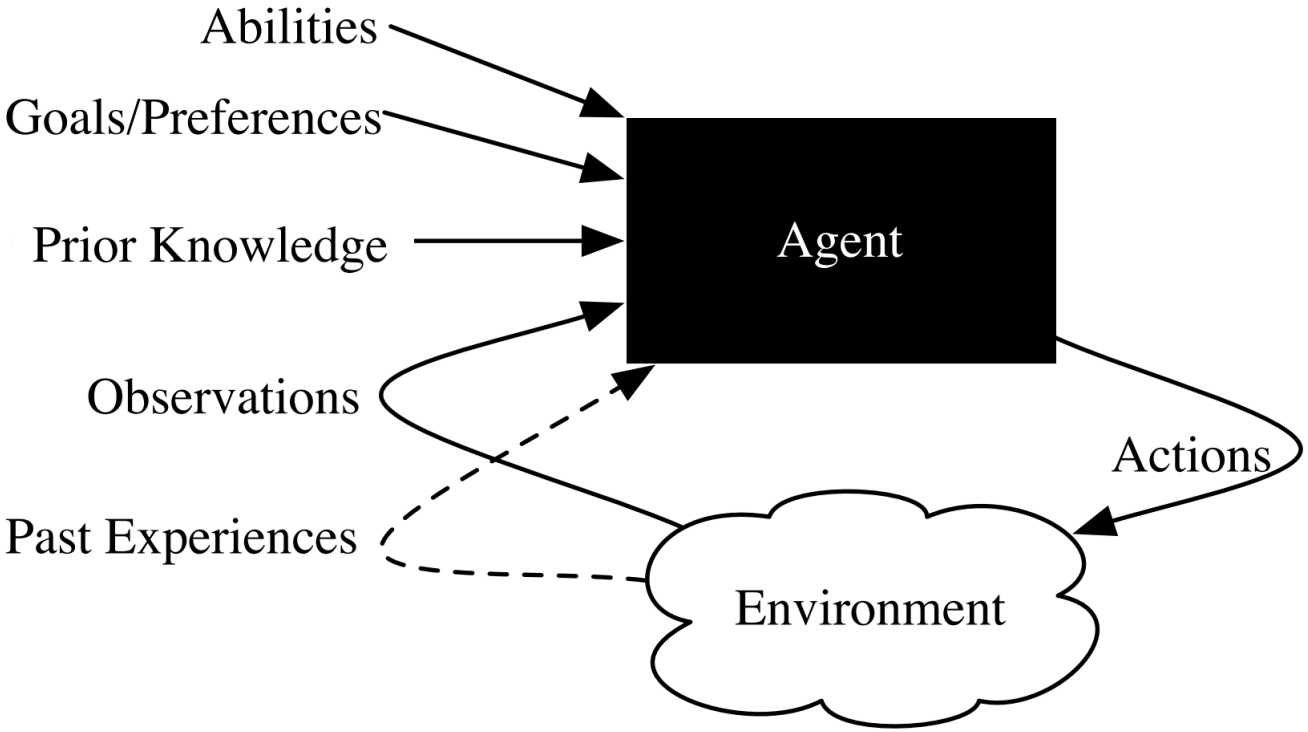
- Resurgence of probability, focus on uncertainty
- General increase in technical depth
- Agents and learning systems… “AI Spring”?

AI in the 1960s

Today’s Agenda

1. What can AI do?
2. What is AI?
3. **AI and Machine Learning**

Artificial Intelligence



Machine Learning

A “machine” that is able to improve its performance in certain tasks based on past experience without explicit human programming on how to improve each time.

Experience: Data
Task: Prediction (classification, recommendation, etc.)
Performance: Accuracy (etc.)

Machine Learning Model

Features X
Target y

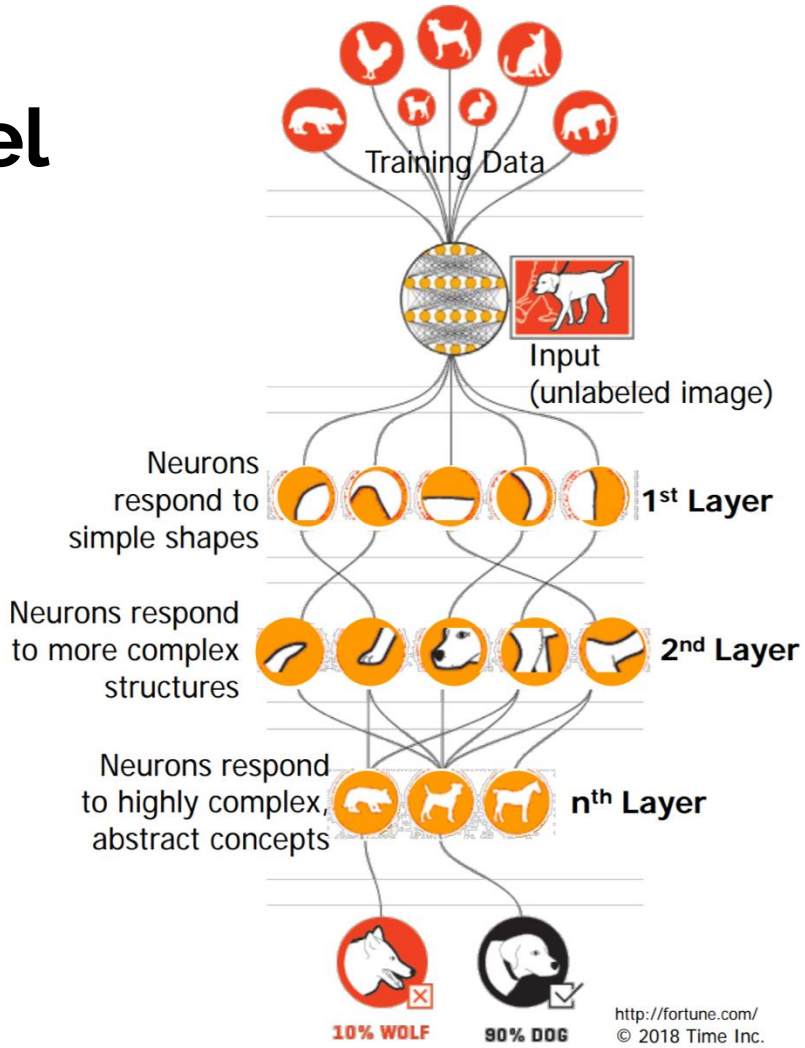
Training Data
(experience)

Machine Learning
Algorithm
(task)

New Example
X_{new}

Model

??
$y_{predicted}$



Machine Learning Tasks

Supervised learning

- Classification
  - Is this a dog or a cat?
- Regression
  - How warm will it be tomorrow?

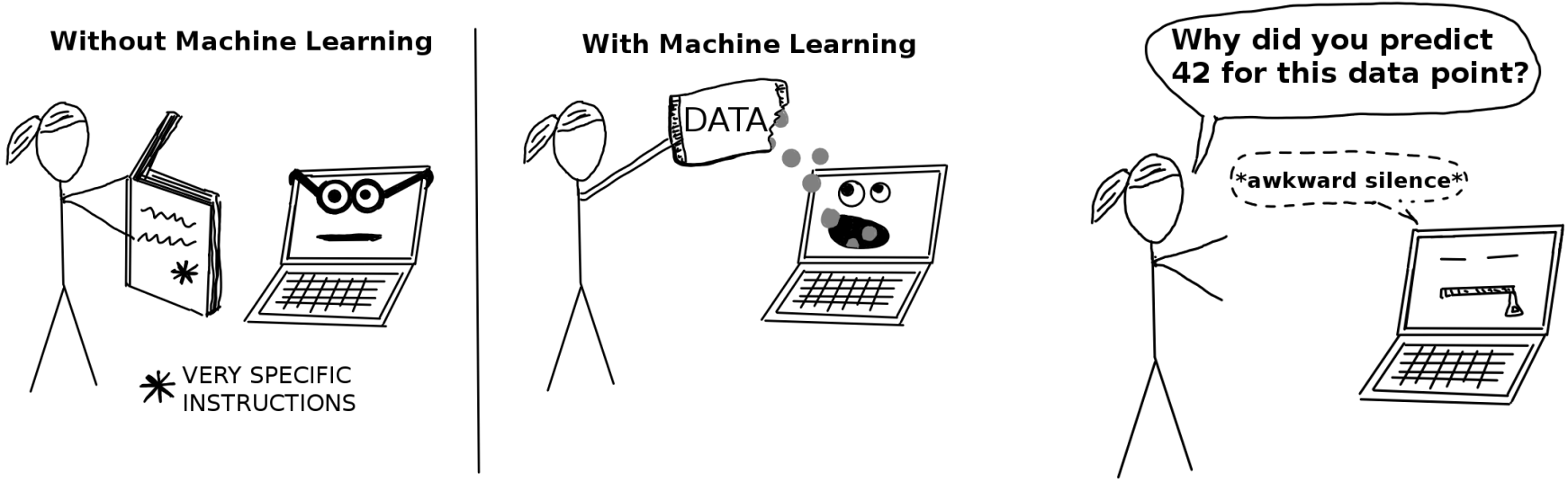
Unsupervised learning

- Clustering
  - Organize a few articles (without labels) into topics
- Representation and embeddings
  - Find a way to represent words, so that representation of “autumn” and “fall” are similar

Reinforcement learning

- Make decisions on actions based on expected reward
  - Drive a car
  - Play Go or chess

The Black-box Problem



https://christophm.github.io/interpretable-ml-book/index.html